

OBJECTIVE

This report summarizes the Untitled.ipynb notebook in the gridsearchcv project. The notebook demonstrates how to tune machine learning hyperparameters using cross-validation, GridSearchCV, and RandomizedSearchCV on the Iris dataset.

DATASET

- Dataset: Iris flower classification dataset from scikit-learn.
- Samples: 150
- Features: 4
- Classes: setosa, versicolor, virginica

Example rows used in the notebook:

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	flower
4.6000	3.2000	1.4000	0.2000	setosa
5.3000	3.7000	1.5000	0.2000	setosa
5.0000	3.3000	1.4000	0.2000	setosa
7.0000	3.2000	4.7000	1.4000	versicolor
6.4000	3.2000	4.5000	1.5000	versicolor

BASELINE MODEL

The notebook first trains an SVM classifier with an RBF kernel, C=30, and gamma='auto' using a 70/30 train-test split.

- Test split accuracy: 1.0000

CROSS-VALIDATION CHECKS

The notebook compares several SVM configurations using 5-fold cross-validation.

kernel	C	fold_scores	mean_score
linear	10	1.000, 1.000, 0.900, 0.967, 1.000	0.9733
rbf	10	0.967, 1.000, 0.967, 0.967, 1.000	0.9800
rbf	20	0.967, 1.000, 0.900, 0.967, 1.000	0.9667

MANUAL HYPERPARAMETER SEARCH

Before using GridSearchCV, the notebook manually loops over SVM kernels and C values and averages 5-fold cross-validation scores.

kernel	C	mean_score
rbf	1	0.9800
rbf	10	0.9800
linear	1	0.9800
linear	10	0.9733
rbf	20	0.9667
linear	20	0.9667

GRIDSEARCHCV RESULTS

GridSearchCV evaluates all combinations of C and kernel for the SVM classifier.

param_C	param_kernel	mean_test_score	rank_test_score
1	linear	0.9800	1
1	rbf	0.9800	1
10	rbf	0.9800	1
10	linear	0.9733	4
20	rbf	0.9667	5
20	linear	0.9667	6

Best GridSearchCV result:

- Best score: 0.9800
- Best parameters: {'C': 1, 'kernel': 'rbf'}

RANDOMIZEDSEARCHCV RESULTS

RandomizedSearchCV samples two parameter combinations from the same SVM search space.

param_C	param_kernel	mean_test_score	rank_test_score
1	rbf	0.9800	1
1	linear	0.9800	1

MODEL COMPARISON

The notebook also compares SVM, Random Forest, and Logistic Regression using GridSearchCV for each model family.

model	best_score	best_params
svm	0.9800	{'C': 1, 'kernel': 'rbf'}
logistic_regression	0.9800	{'C': 5}
random_forest	0.9667	{'n_estimators': 10}

CONCLUSION

- Grid search provides a systematic way to evaluate every configured hyperparameter combination.
- For this Iris dataset, SVM configurations with $C=1$ tied for the strongest cross-validation score, with GridSearchCV selecting the RBF kernel.
- Randomized search is useful when the search space is large, but with only two sampled configurations it may miss the best available option.
- Cross-validation gives a more reliable estimate than relying on one train-test split.